Next, let $MI' = \frac{MI}{\left(\hat{b}_1 POV + \left(1 - \hat{b}_1\right)\right)}$ then we have:

$$MI' = I^*$$

Now rewrite I^* in its full functional form:

$$MI' = a_1 \cdot \left(\frac{Q}{ADV}\right)^{a_2} \cdot \sigma^{a_3}$$

The model parameters can now be estimated via non-linear regression techniques such as non-linear least squares or maximum likelihood estimates. It is important for analysts to evaluate heteroscedasticity and multicollinearity across the data.

After grouping the data, we often see the equation's LHS MI' will have all positive values (or only a few categories with negative values). In these cases we can perform a log transformation of the data. If all LHS data points are positive then we can transform all data points. If there are a few grouping categories with negative values we can transform all the positive LHS records and either eliminate the groupings with negative values or give these records a transformed LHS value of say -3 or -5 (the exact value will depend upon the positive LHS values and units). Another benefit of performing a log transformation is that this transformation will correct (approximately) for the heteroscedasticity—no other adjustment is required.

The log transformation of the equation is as follows:

$$ln(MI') = ln(\alpha_1) + \alpha_2 ln\left(\frac{Q}{ADV}\right) + \alpha_3 ln(\sigma)$$

This regression can now be solved via OLS—which is nice and direct and has easy statistics to interpret. The estimated I-Star parameters are then:

$$a_1 = exp(\alpha_1 + 0.5 \cdot SE),\ a_2 = \alpha_2, \text{ and } a_3 = \alpha_3$$

It is important to note that the two-step process can also be performed using the a_4 parameter and non-linear temporary market impact rate. In this case, we would estimate values for b_1 and a_4 in the first steps, and proceed in exactly the same manner in the second step starting with:

$$MI' = \frac{MI}{\left(\hat{b}_1 POV^{a_4} + \left(1 - \hat{b}_1\right)\right)}$$